

Programme Specification

Title of Course: MEng Mechanical Engineering

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This Programme Specification is designed for prospective students, current students, academic staff and potential employers. It provides a concise summary of the main features of the programme and the intended learning outcomes that a typical student might reasonably be expected to achieve and demonstrate if he/ she takes full advantage of the learning opportunities that are provided. More detailed information on the teaching, learning and assessment methods, learning outcomes and content of each module can be found in Student Handbooks and Module Descriptors.

SECTION 1: GENERAL INFORMATION

Title:	MEng Mechanical Engineering
Awarding Institution:	Kingston University
Teaching Institution:	Kingston University
Location:	Roehampton Vale Campus, Kingston
Programme Accredited by:	Institution of Mechanical Engineers (IMechE)

SECTION 2: THE PROGRAMME

A. Programme Introduction

The Programme is aimed at preparing students who aim to work in Engineering and aspire to achieve their professional status of Chartered Engineer (CEng). The Programme embraces recent developments in education and industry. The Programme design is based on the guidelines provided by the Engineering Council UK Standard for Professional Engineering Competence (UK- SPEC), the Quality Assurance Agency (QAA) Subject Benchmark Statement for Engineering, and the Institution of Mechanical Engineers (IMechE) Academic Accreditation Guidelines.

MEng graduates should have the versatility and depth of understanding to enable them to deal with new and unusual challenges in their chosen field of engineering. They should be imaginative and creative so that they can become innovators. MEng graduates must be equipped to progress rapidly to a position of responsibility and provide technical, managerial, and entrepreneurial leadership in specialist or inter-disciplinary projects.

The programme has a reputation for academic excellence, intellectual rigour and industrial relevance and is designed to equip graduates with the knowledge, comprehension, intellectual ability and subject practical skills to become professional mechanical engineers or to follow careers in related professional areas. Employability is a key element of the Programme and hence the emphasis on communication, interpersonal and other skills that today's industry sees as enhancing employment prospects.

The first two years, levels four and five, of the MEng provides a broad foundation in Mechanical Engineering appropriate to the aims, objectives and learning outcomes of the programme. All MEng graduates require a foundation that covers the broad spectrum of engineering and this is provided for by studying topic which go beyond specialisation in Mechanical Engineering, hence professional issues, engineering practice and topics such as computing and electrical and electronic systems are all introduced here. In the final two years, levels six and seven, the Programme deals in much more depth with those topics which are more specific to a specialisation in Mechanical Engineering and encourages independence through the use of individual

and group project work, allowing students to showcase their abilities to deal with complex and unfamiliar data, problems and situations.

The Programme broadly follows four themes or threads, and these threads are built upon from level to level. Firstly the appreciation and application of the principles of engineering science through the core knowledge deepening subject areas of mechanical systems (statics and dynamics), thermodynamics and fluid mechanics are emphasised. Secondly, topics intended to widen the students' knowledge base include electrical and electronic systems, control engineering, mechatronics and computing. Thirdly there is a professional theme, introducing the students to the practice of a Mechanical Engineer, covering such topics as professional practice, project management, quality and business management. Lastly there is a strong design theme linking together the other threads and emphasising the holistic nature of modern day engineering. Hence, the MEng provides both breadth and depth with an aim to develop the ability to identify, define and solve engineering problems from first principles.

This MEng course emphasises the development of practical skills and experimentation through the extensive use of modern well equipped laboratories and workshops. Sustainability and 'Health and Safety' are threaded throughout the Programme's modules and students are encouraged to consider the impact that engineering has on their environment. There is a great deal of support available to students, both pastorally and academically, but in particular they are supported by a Personal Tutor Scheme (PTS) in which they are allocated a member of staff who, through one-to-one meetings, will assist and encourage students in their academic learning for the duration of their Course.

The MEng is offered as a four-year full-time degree course or a five-year sandwich course with an industrial placement taken between level 5 and level 6, or between levels 6 and 7. The industrial placement is strongly recommended and supported as the experience gained goes a long way in preparing students for employment and, along with the academic experience gained at Kingston University, ensures that our graduates are widely recognised as being well prepared for the world of work. Many of our graduates from the previous programme arrangement have gone on to very successful careers.

B. Aims of the Programme

The general aims of the course are:

- To equip graduates with the engineering, design, management, business and personal skills required to become professional Mechanical Engineers, as well as enabling them to follow careers in related professional disciplines.
- To fully meet the exemplifying academic benchmark requirements, for registration as a Chartered Engineer (CEng) and for Chartered Membership of the Institution of Mechanical Engineers (IMechE), by ensuring that the course is accredited by that body.